

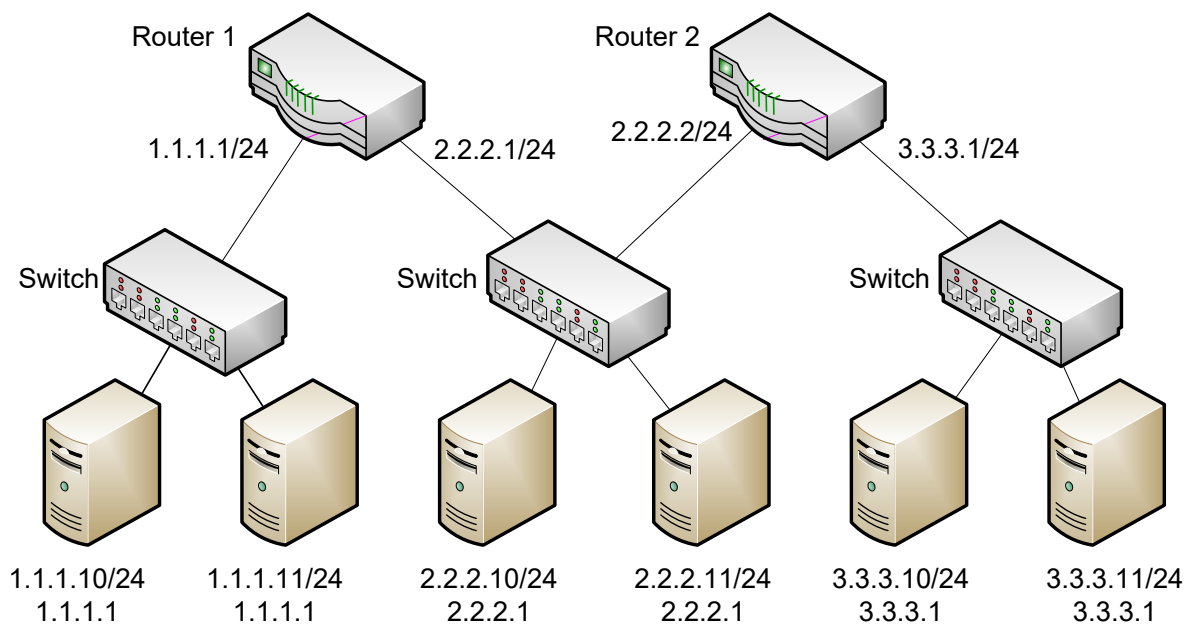
Lab 11

Routing Protocol Operation

Objective:

The purpose of this lab is for the student to understand routing protocol functionality. Students will also utilize IOS commands to explore routing protocol functionality and features.

Diagram:



Components:

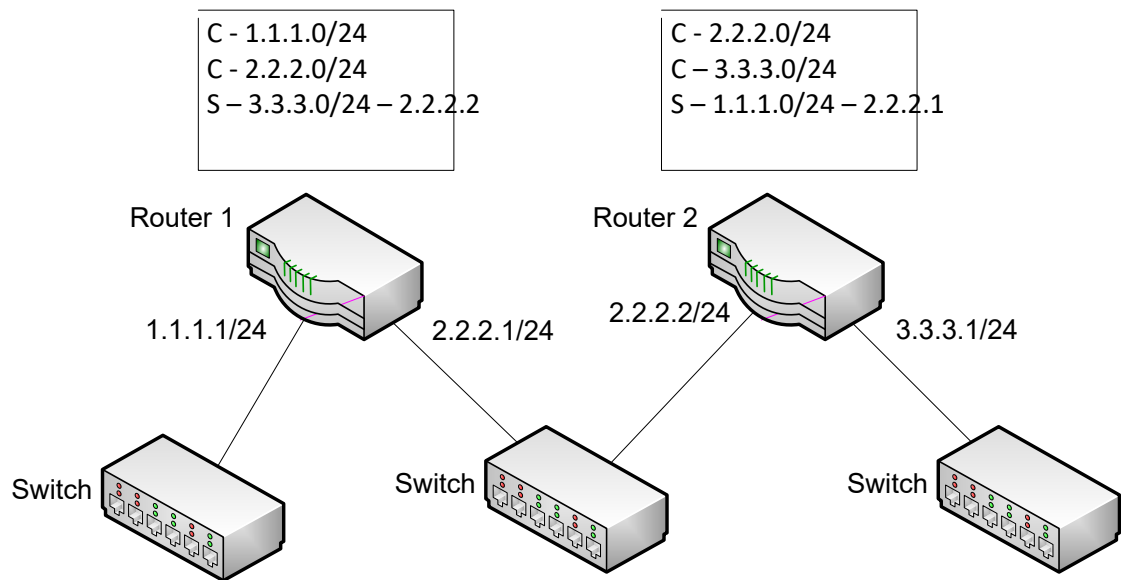
6 – PC's with an Ethernet NIC

3 – Switches

2 – 2 Port Routers

Procedure:

1. Connect all PC's together and put their static IP addresses, subnet masks and gateways on them. Test connectivity by pinging each machine within the network.
2. On the diagram, label the routing tables below.



3. Enable the router interfaces and place IP addresses on them.
4. View Router 1's routing table.
 - a. What routes are displayed?
 - 1.1.1.0/8 is variably subnetted, 2 subnets, 2 masks
 - C 1.1.1.0/24 is directly connected, GigabitEthernet0/0/0
 - L 1.1.1.1/32 is directly connected, GigabitEthernet0/0/0
 - 2.2.2.0/8 is variably subnetted, 2 subnets, 2 masks
 - C 2.2.2.0/24 is directly connected, GigabitEthernet0/0/1
 - L 2.2.2.1/32 is directly connected, GigabitEthernet0/0/1
 - b. What routes are missing? Network 3

5. View Router 2's routing table.

a. What routes are displayed?

2.2.2.0/8 is variably subnetted, 2 subnets, 2 masks

C 2.2.2.0/24 is directly connected, GigabitEthernet0/0/0

L 2.2.2.2/32 is directly connected, GigabitEthernet0/0/0

3.3.3.0/8 is variably subnetted, 2 subnets, 2 masks

C 3.3.3.0/24 is directly connected, GigabitEthernet0/0/1

L 3.3.3.1/32 is directly connected, GigabitEthernet0/0/1

b. What routes are missing? Network 1

6. Enable the EIGRP routing protocol on router 1 with an AS of 1

a. What commands did you enter?

Router eigrp 1

Network 1.1.1.0 0.0.0.255

Network 2.2.2.0 0.0.0.255

b. View the routing table. Are the missing routes there? No

c. Why? The second router has not been configured yet

7. Enable the EIGRP routing protocol on router 2 with an AS of 1

a. What commands did you enter?

Router eigrp 1

Network 2.2.2.0 0.0.0.255

Network 3.3.3.0 0.0.0.255

b. View the routing table. Are the missing routes there? Yes

c. Why? The routers have exchanged their EIGRP information now

8. View the routing table on router 1. Are the missing routes there? Yes

a. Why? The routers have exchanged their EIGRP information now

9. Can 1.1.1.10 ping 3.3.3.10? Yes

10. Change the AS number on Router 1 to 2. View the routing table.

a. What happened to the routing table? It dropped the 3 network

b. Can 1.1.1.10 ping 3.3.3.10? No

c. Why? By changing the rule number, the routers no longer know how to communicate with each other to find the path to the other networks.

11. Remove the EIGRP routing protocols on Router 1 and 2

a. What commands did you enter?

No router eigrp 2

No router eigrp 1

12. View the routing table. Are the routes gone? yes

13. Enable the OSPF routing protocol on router 1 with Areas of 0 and a process ID of 1

a. What commands did you enter?

Router ospf 1

Network 1.1.1.0 0.0.0.255 area 0

Network 2.2.2.0 0.0.0.255 area 0

b. View the routing table. Are the missing routes there? No

c. Why? The ospf has not been configured on router 2 yet.

14.Enable the OSPF routing protocol on router 2 with Areas of 0 and a process ID of 1.

a. What commands did you enter?

Router ospf 1

Network 2.2.2.0 0.0.0.255 area 0

Network 3.3.3.0 0.0.0.255 area 0

b. View the routing table. Are the missing routes there? Yes

c. Why? The routers have shared ospf information now

15.View the routing table on router 1. Are the missing routes there? Yes

a. Why? The routers have shared ospf information now

16.Can 1.1.1.10 ping 3.3.3.10? yes

17.Change the Area numbers on Router 1 to 2. View the routing table.

a. What happened to the routing table? It dropped network 3

b. Can 1.1.1.10 ping 3.3.3.10? no

c. Why? Because it no longer knows the path to network 3 since the area numbers were changed

18.Remove the OSPF routing protocols on Router 1 and 2

a. What commands did you enter?

No router ospf 2

No router ospf 1

19.View the routing table. Are the routes gone? Yes

a. Why? The rule was deleted, so it does not know how to route outside of its LAN.

20.Enable static routing on Router 1

a. What commands did you enter?

Ip route 3.3.3.0 255.255.255.0 2.2.2.2

b. View the routing table. Are the missing routes there? Yes

c. Why? Because the static route was configured, and it shows up immediately

21.Enable the static routing on Router 2

a. What commands did you enter?

Ip route 1.1.1.0 255.255.255.0 2.2.2.1

b. View the routing table. Are the missing routes there? Yes

c. Can 1.1.1.10 ping 3.3.3.10? Yes

d. Why? Both routers are configured with static routes.

22.Remove the static routes on Router 1 and 2

a. What commands did you enter?

No ip route 3.3.3.0 255.255.255.0 2.2.2.2

No ip route 1.1.1.0 255.255.255.0 2.2.2.1

23. View the routing table.

a. Are the routes gone? Yes

b. Why? Because the static routes were deleted

c. Can 1.1.1.10 ping 3.3.3.10? No

Draw the lab diagram below (PC's, switch, and router connections) Label the IP addresses, and subnet masks of each device. Feel free to use packet tracer or Visio, but do not copy and paste the diagram at the beginning of the lab. Screen shot your diagram and paste it below. Label the routing tables next to the routers.

